

Vulnerability Assessment Report

6 August 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business?*
- *Why is it important for the business to secure the data on the server?*
- *How might the server impact the business if it were disabled?*

The purpose of this vulnerability assessment is to evaluate the security risks associated with the company’s database server. The database server is valuable because it stores customer information and supports important business operations. Securing the data is important to protect confidentiality, integrity, and availability. If the server were disabled or compromised, it could disrupt business operations, cause financial losses, and expose sensitive customer information.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
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<i>External attackers</i>	<i>Unauthorised access to customer data</i>	3	3	9
<i>Cybercriminals</i>	<i>Data theft through database exploitation</i>	2	3	6
<i>Malicious insiders</i>	<i>Misuse of database access privileges</i>	2	2	4

Approach

This vulnerability assessment uses a qualitative risk assessment approach to evaluate the likelihood and severity of potential threats. The selected threats were chosen because the database server is publicly accessible and contains sensitive customer information. External attackers, cybercriminals, and malicious insiders could impact the confidentiality, integrity, and availability of the data. These risks are significant because successful attacks could cause data breaches, financial losses, and disruption to business operations.

Remediation Strategy

To remediate the identified risks, the organization should remove public access to the database server and implement access controls. The company should apply the principle of least privilege and use multi-factor authentication to prevent unauthorized access. Regular vulnerability scans, security updates, and system monitoring should be performed to identify and address security weaknesses. These defense-in-depth strategies will help protect sensitive customer data and reduce the likelihood of successful attacks.